

Exercise 1.1.2. Prove this statement: if $\mathcal{F}$ is a collection of subsets which is closed under complement and countable unions, then it is closed under countable intersections.

1.1.2. proof. Let $(A_i)_{i \in \mathbb{N}}$ be a sequence in $\mathcal{F}$. So $(A_i^c)_{i \in \mathbb{N}}$ is also a sequence in $\mathcal{F}$ since it is closed under complement. So $\bigcup_i A_i^c \in \mathcal{F}$. By de Morgan's law, $(\bigcup_i A_i^c)^c = \bigcap_i A_i$. so $\bigcap_i A_i \in \mathcal{F}$ $\square$

Exercise 1.1.3. If Ω contains N elements, how many elements does its power set contain? Provide an easy and understandable description of this power set.

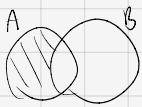
1.1.3 For each elements, we have 2 choices, yes or no, to include it in a set. so $|\mathcal{P}(\Omega)| = 2^N$

Exercise 1.1.6. If $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space and if $A, B \in \mathcal{F}$, check that

- 1) $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$, where $A^c := \Omega \setminus A$,
- 2) $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$,
- 3) If $A \subset B$, then $\mathbb{P}(A) \leq \mathbb{P}(B)$.

1.1.6.1) $\mathbb{P}(\Omega) = \mathbb{P}(A \cup A^c) = \mathbb{P}(A) + \mathbb{P}(A^c) \Rightarrow \mathbb{P}(A^c) = \mathbb{P}(\Omega) - \mathbb{P}(A) = 1 - \mathbb{P}(A)$

2) $A \cup B = (A \setminus B) \cup B$ $\mathbb{P}(A \cup B) = \mathbb{P}(A \setminus B) + \mathbb{P}(B) = \mathbb{P}(A) - \mathbb{P}(A \cap B) + \mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$



$A \setminus B = (A \cap B^c) \cup (A \cap B)$ $\Rightarrow \mathbb{P}(A \setminus B) = \mathbb{P}(A) - \mathbb{P}(A \cap B)$

3) suppose $A \subset B$ & by contradiction $\mathbb{P}(A) > \mathbb{P}(B)$. we see that $B = (B \setminus A) \cup A$ so $\mathbb{P}(B) = \mathbb{P}(B \setminus A) + \mathbb{P}(A) \Rightarrow$

$\mathbb{P}(B) \geq \mathbb{P}(A)$ since $\mathbb{P}$ is a non-negative function.

Lemma 1.1.7 (Continuity of probability). Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and consider $\{A_j\}_{j \in \mathbb{N}} \subset \mathcal{F}$. If $A_j \subset A_{j+1}$ for any j , then

$$\mathbb{P}\left(\bigcup_{j \in \mathbb{N}} A_j\right) = \lim_{j \rightarrow \infty} \mathbb{P}(A_j),$$

while if $A_j \supset A_{j+1}$ for any j , then

$$\mathbb{P}\left(\bigcap_{j \in \mathbb{N}} A_j\right) = \lim_{j \rightarrow \infty} \mathbb{P}(A_j).$$

The proof is left as an exercise, or can be found in [1, Lem. 1.4].

1.1.7 $(\Omega, \mathcal{F})$ be a measurable space and f be a finite measure. Then $f(\bigcup A_j) = \lim f(A_j)$ for $A_j \subset A_{j+1}$, and

$f(\bigcap A_j) = \lim f(A_j)$ for $A_j \supset A_{j+1}$. Instead of just $\mathbb{P}$, we generalize it to finite measure.

proof. ver 1 for union $f(A_j)$ is increasing and bounded above since f is finite, so $\lim_{j \rightarrow \infty} f(A_j)$ exists.

(uses monotone sequence property, Least upper bound property, but require finiteness of measure) clearly, $\forall j, \bigcup A_j \supset A_j$. So $\forall j, f(\bigcup A_j) \geq f(A_j)$. Therefore $f(\bigcup A_j) \geq \lim_{j \rightarrow \infty} f(A_j) = \lim_{j \rightarrow \infty} \sup_{m \geq j} f(A_m)$.

Next we want to show $\lim_{j \rightarrow \infty} \sup_{m \geq j} f(A_m) \leq f(\bigcup A_j)$. We can do this by showing $\forall j, \sup_{m \geq j} f(A_m) \leq f(\bigcup A_j)$.

By contradiction, $\exists j$ s.t. $\sup_{m \geq j} f(A_m) < f(\bigcup A_j)$. Let $M = \{m \in \mathbb{N} \mid \sup_{n \geq m} f(A_n) < f(\bigcup A_j)\} \neq \emptyset$, then

by the greatest lower bound property (as $\mathbb{N}$ is partially ordered), $L := \inf M$ exists.

If $L \notin M$ then we get a contradiction as L would not be the greatest lower bound. so $L \in M$,

therefore $\sup_{m \geq L} f(A_m) < f(\bigcup A_j)$ but $\sup_{m \geq L-1} f(A_m) \geq f(\bigcup A_j)$. since $\sup_{m \geq L-1} f(A_m) = \max\{\sup_{m \geq L} f(A_m), f(A_{L-1})\} = f(A_{L-1})$, we get $\sup_{m \geq L} f(A_m) < f(A_{L-1}) \Rightarrow f(A_L) < f(A_{L-1})$, this is a contradiction since

$A_{i-1} \in A_i$. So we get $\forall j, \sup_{m \geq j} f(A_m) \geq f(\cup A_i)$ which implies $\limsup_{j \rightarrow \infty} \sup_{m \geq j} f(A_m) \geq f(\cup A_i)$.

So we get $f(\cup A_i) \geq \lim_{j \rightarrow \infty} f(A_j) = \limsup_{j \rightarrow \infty} \sup_{m \geq j} f(A_m) \geq f(\cup A_i) \Rightarrow f(\cup A_i) = \lim_{j \rightarrow \infty} f(A_j)$. Done for union.

proof ver 2.

↑

(use countable
additivity)

Let $B_1 = A_1$ and $B_n = A_n \setminus A_{n-1}$, $n \geq 2$. We see that $A_n = \bigcup_{k=1}^n B_k$ and $\cup A_k = \bigcup_{k=1}^{\infty} B_k$.

$f(A_n) = \sum_{k=1}^n f(B_k)$ and $f(\cup A_k) = \sum_{k=1}^{\infty} f(B_k)$. By def $\lim_{n \rightarrow \infty} \sum_{k=1}^n f(B_k) = \sum_{k=1}^{\infty} f(B_k) \Rightarrow$

$\lim f(A_n) = f(\cup A_k)$. Note that we did not need finiteness of measure.

Let $A_i \supseteq A_{i+1}$ then $A_i^c \subset A_{i+1}^c$. $f((\cap A_i)^c) = f(\Omega) - f(\cap A_i)$ and $f((\cap A_i)^c) = f(\cup A_i^c) =$
 $= \lim f(A_i^c) \Rightarrow f(\cap A_i) = f(\Omega) - \lim f(A_i^c) = \lim (f(\Omega) - f(A_i^c)) = \lim (f(A_i))$. □

Note that we need $f(\Omega) < \infty$ for $\cap A_i$ but not for $\cup A_i$.